

ECON 712: Applied Econometrics

Lecture 14: Fixed Effects & First Differencing

Panel Data and Unobserved Heterogeneity

- Observations: y_{it} , x_{it} for units $i = 1, \dots, N$ and times $t = 1, \dots, T$.
- Goal: identify β in $y_{it} = x_{it}'\beta + u_{it}$ when x_{it} correlates with unobserved individual effects.
- Typical structural form:

$$y_{it} = x_{it}'\beta + \alpha_i + \varepsilon_{it},$$

with α_i time-invariant unit-specific factor.

- If $E[x_{it}\alpha_i] \neq 0$, pooled OLS is inconsistent.
- Fixed effects (FE) and first differencing (FD) remove α_i under weak assumptions.

FE: Subtract Unit Mean

Consider

$$y_{it} = x'_{it}\beta + \alpha_i + \varepsilon_{it}.$$

Define time averages:

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{x}_i = \frac{1}{T} \sum_{t=1}^T x_{it},$$

$$\bar{\varepsilon}_i = \frac{1}{T} \sum_{t=1}^T \varepsilon_{it}.$$

Subtract:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

- α_i drops out cleanly.
- OLS on transformed data gives FE estimator:

$$\hat{\beta}_{FE} = \left(\sum_{i,t} \tilde{x}_{it} \tilde{x}'_{it} \right)^{-1} \left(\sum_{i,t} \tilde{x}_{it} \tilde{y}_{it} \right),$$

with $\tilde{z}_{it} = z_{it} - \bar{z}_i$.

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FD: Differences across Time

For $T \geq 2$, take first differences:

$$\Delta y_{it} = y_{it} - y_{i,t-1},$$

$$\Delta x_{it} = x_{it} - x_{i,t-1},$$

$$\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}.$$

Then

$$\Delta y_{it} = \Delta x'_{it} \beta + \Delta \varepsilon_{it},$$

and α_i is eliminated.

- FD estimator is OLS on differenced data for $t = 2, \dots, T$.
- Equivalent to FE with $T = 2$; in general both estimate same β when regressors are strictly exogenous.

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FE = De-meaning (within) = PDV

- FE transformation: $M_i = I_T - (1/T)1_T1_T'$ on each unit block.
- FD transformation: D_i with $D_i y_i = (y_{i2} - y_{i1}, \dots, y_{iT} - y_{i,T-1})'$.
- Both remove α_i and deliver identical β under $E[x_{it}\varepsilon_{is}] = 0$ for all t, s .
- If ε_{it} serially uncorrelated and x_{it} strictly exogenous, FE and FD are asymptotically equivalent.
- Practical implication: use FE as default when T moderate and serial correlation is possible.

Incidental Parameters Problem

- Neyman-Scott (1948): incidental parameters can bias MLE when T fixed and $N \rightarrow \infty$.
- In linear FE, α_i estimated as $(1/T) \sum_t (y_{it} - x'_{it} \hat{\beta})$, but we do not care about $\hat{\alpha}_i$ for β consistency.
- FE estimator solves the same normal equations as within-transformed OLS, and $\hat{\beta}_{FE}$ is consistent for fixed T as $N \rightarrow \infty$.
- Incidental parameter bias is absent because linear model with homoskedastic errors has exact FE elimination, not nonlinear likelihood.
- With T finite, estimate of $Var(\varepsilon_{it})$ needs degrees-of-freedom correction $(N(T-1) - K)$, but β remains consistent.

Serial Correlation in Panel PP Errors

- If ε_{it} is autocorrelated within i (e.g., AR(1)), then

$$E[\tilde{x}_{it}\tilde{\varepsilon}_{is}] \neq 0 \text{ for } t \neq s$$

can violate strict exogeneity unless x_{it} predetermined.

- In FE and FD, standard errors must allow within-cluster correlation.
- Use clustered standard errors by i :

$$\widehat{Var}(\hat{\beta})_{cluster} = (X'X)^{-1} \left(\sum_i X_i' \hat{u}_i \hat{u}_i' X_i \right) (X'X)^{-1}.$$

- Clustering adjusts for arbitrary T -dependence, including serial correlation and heteroskedasticity across i .

Practical Tips

- Always report number of clusters (units) and time periods.
- If T small (e.g., 3-5), use wild cluster bootstrap or t-statistics with CCE.
- Compare FE to random effects (RE) with Hausman test when α_i uncorrelated with x_{it} .
- In nonlinear models (logit/probit), incidental parameters require other estimators (conditional MLE, bias correction), but linear FE is safe for β .
- Use lagged regressors carefully: $x_{i,t-1}$ is valid if you assume weak exogeneity rather than strict exogeneity.

References

- Wooldridge (2010) ch. 10: fixed effects and first differences.
- Angrist & Pischke (2014): Section on panel methods, two-way FE.
- Causal Inference: The Mixtape (Kleinberg et al.) ch. 10: panel data, FE and change scores.
- "ECON 712 Slides - February 27th 2 (W23)" [class slides].